

Parent Functions III: $1/x$ and Asymptotes**TEACHER KEY** | Algebra II | Unit 1 | Day 6 | TEK A2.2.A**Objective:** I can graph the reciprocal parent function, identify its vertical and horizontal asymptotes, and describe its domain, range, and symmetry.**Vocabulary****Asymptotic behavior** — how a graph behaves near a value it gets closer and closer to but never **reaches**.**Vertical asymptote** — a vertical line the graph **never crosses**. It sits where the function is undefined.**Horizontal asymptote** — a horizontal line the y-values get **closer and closer to, without touching**, as x gets very large or very small.**Part 1 — Build the Table for $f(x) = 1/x$**

Fill in the outputs. One input has no output at all.

x	-2	-1	-1/2	0	1/2	1	2
$f(x) = 1/x$	-1/2	-1	-2	undef.	2	1	1/2

Why is $x = 0$ undefined? **you cannot divide by zero****Part 2 — What Happens Near Zero and Far Out**

Follow the pattern in each list, then describe it.

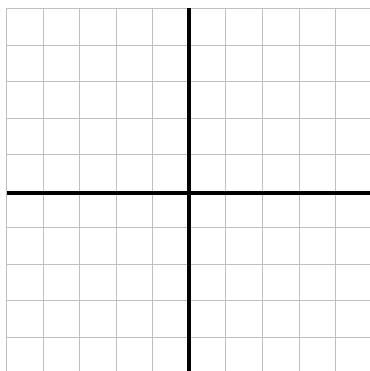
x	1	0.5	0.1	0.01
$1/x$	1	2	10	100

As x gets close to 0, the outputs grow **without bound**. So there is a vertical asymptote at **$x = 0$** .

x	1	10	100	1000
$1/x$	1	0.1	0.01	0.001

As x gets very large, the outputs approach **0**. So there is a horizontal asymptote at **$y = 0$** .**Part 3 — Graph It**

Each square is 1 unit. Plot your table points. You will get two separate branches - do not connect them across the y-axis.

 $f(x) = 1/x$ **Record what you see.**Vertical asymptote: **$x = 0$** Horizontal asymptote: **$y = 0$** How many branches? **2**Which quadrants? **I and III**Any intercepts? **none**

Draw both asymptotes as dashed lines.

Part 4 — Key Attributes of $f(x) = 1/x$

Domain: **all reals except 0** set notation $\{x \mid x \in \mathbb{R}, x \neq 0\}$

Range: **all reals except 0** interval $(-\infty, 0) \cup (0, \infty)$

Symmetry: reflectional across **$y = x$ and $y = -x$** , and rotational **180°** about the origin.

Maximum or minimum? **neither** Intercepts? **none**

This is the first parent function with **no intercepts** and the first whose domain has a **hole** in it.

Part 5 — True or False

1. A graph may cross a vertical asymptote. **false**
2. A graph may cross a horizontal asymptote. **true**
3. $f(x) = 1/x$ has a y-intercept. **false**
4. The domain of $1/x$ is every real number. **false**

Practice

Evaluate and explain. Show your work where there is room.

1. Evaluate: $1/4 = \mathbf{0.25}$ $1/(-2) = \mathbf{-0.5}$ $1/(1/5) = \mathbf{5}$
2. Solve $1/x = 4$. $x = \mathbf{1/4}$ Solve $1/x = -1$. $x = \mathbf{-1}$
3. Is there any x with $1/x = 0$? Explain.

4. As x grows very large and negative, what happens to $1/x$? **it approaches 0 from below**
5. Fill in the pattern.

x	-0.1	-0.01	0.01	0.1
1/x	-10	-100	100	10

Approaching 0 from the left the outputs go toward **$-\infty$** , from the right toward **$+\infty$** .

6. Why is $1/x$ the only parent function so far with a gap in its domain?

Exit Ticket

A function has no intercepts, a vertical asymptote at $x = 0$, a horizontal asymptote at $y = 0$, and two branches in Quadrants I and III.

Name it: **$f(x) = 1/x$** Domain in set notation: **$\{x \mid x \in \mathbb{R}, x \neq 0\}$**